

Generation Interconnection System Impact Study Report

For Queue Project Q12-0163

James E. Rogers Energy Complex 138 kV
October 21, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0163. The Interconnection Customer proposes a 246.0 MW wind facility interconnecting at the James E. Rogers Energy Complex 138 kV substation in the Pennsylvania area. The request was received on October 08, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 103 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$35,969,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

RAINTREE - BAKEOVEN 138 kV: 159.7% loading (198 MVA) following the outage of the N ALLEGHENY WINDPOWER PROJECT - BAKEOVEN 138 kV line; DFAX = 37.1%.

For the loss of BAKEOVEN - BEARDSLEY 138 kV, the NOTTELY - MARION S 138 kV loads to 141.6% of its 209 MVA rating. Project DFAX: 20.9%.

For the loss of BAKEOVEN - BEARDSLEY 138 kV, the NOTTELY - 7TH AVENUE N 138 kV loads to 140.1% of its 213 MVA rating. Project DFAX: 20.9%.

RAINTREE - MUNISING 138 kV L: 138.3% loading (221 MVA) following the outage of the BAKEOVEN - NORTH ALLEGHENY WINDPOWER PROJECT 138 kV; DFAX = 37.1%.

CARL BAILEY - OTSEGO SUBSTATION 138 kV: 126.4% loading (400 MVA) following the outage of the NORWAY PT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138 kV; DFAX = 6.2%.

CARREL 345-138 kV Autotransformer: 123.0% loading (490 MVA) following the outage of the FAIR OAKS - CHAMBERSBURG 345 kV line; DFAX = 7.0%.

For the loss of SANTEE COOPER - GREENLAND GAP 345 kV, the SPINDLE 500/345kV Auto loads to 121.6% of its 750 MVA rating. Project DFAX: 5.5%.

HIGHLAND WIND PROJECT PA - KENNECOTT SUBSTATION 138 kV line: 117.7% loading (422 MVA) following the outage of the NORWAY PT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138 kV Line; DFAX = 54.6%.

The ALCONA - HIGHLAND WIND PROJECT PA 138 kV is overloaded to 117.6% of its 424 MVA emergency rating for the loss of the JAMES E. ROGERS ENERGY COMPLEX - NORWAY PT HYDROPOWER PROJECT 138 kV. The project contributes 54.6% DFAX (134.4 MW) to this overload.

The BARNUM - ALCONA 138 kV is overloaded to 117.2% of its 430 MVA emergency rating for the loss of the JAMES E. ROGERS ENERGY COMPLEX - NORWAY POINT HYDROPOWER PROJECT 138 kV. The project contributes 54.6% DFAX (134.4 MW) to this overload.

For the loss of NORWAY POINT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138 kV, the BARNUM - WOLFSKILL TAP 138 kV Line loads to 116.7% of its 436 MVA rating. Project DFAX: 54.6%.

For the loss of BEARDSLEY SUBSTATION - BAKEOVEN 138 kV, the MANCHACA - O'BANION 138 kV loads to 116.5% of its 348 MVA rating. Project DFAX: 40.2%.

MERRILL HILLS - WOLFSKILL TAP 138 kV line: 116.5% loading (439 MVA) following the outage of the NORWAY PT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138 kV; DFAX = 54.6%.

The MERRILL HILLS - COLLIERVILLE 138 kV is overloaded to 116.3% of its 442 MVA emergency rating for the loss of the JAMES E. ROGERS ENERGY COMPLEX - NORWAY PT HYDROPOWER PROJECT 138 kV. The project contributes 54.6% DFAX (134.4 MW) to this overload.

O'BANION - PURDUE NORTHWEST SUBSTATION 138 kV: 116.0% loading (354 MVA) following the outage of the BEARDSLEY - BAKEOVEN 138 kV; DFAX = 40.2%.

The BEARDSLEY - PURDUE NORTHWEST 138 kV is overloaded to 115.7% of its 358 MVA emergency rating for the loss of the BEARDSLEY - BAKEOVEN 138 kV. The project contributes 40.2% DFAX (99.0 MW) to this overload.

The BAKEOVEN - NORTH ALLEGHENY WINDPOWER PROJECT 138 kV L is overloaded to 114.1% of its 454 MVA emergency rating for the loss of the JAMES E. ROGERS ENERGY COMPLEX SUBSTATION - COLLIERVILLE 138 kV. The project contributes 51.5% DFAX (126.7 MW) to this overload.

BEARDSLEY - BAKEOVEN 138 kV: 113.7% loading (642 MVA) following the outage of the JAMES E. ROGERS ENERGY COMPLEX - COLLIERVILLE 138 kV; DFAX = 72.0%.

For the loss of BEARDSLEY SUBSTATION - BAKEOVEN 138 kV L, the 7TH AVENUE N - CIMGEN 138 kV Line loads to 113.6% of its 386 MVA rating. Project DFAX: 40.2%.

For the loss of BAKEOVEN - BEARDSLEY SUBSTATION 138 kV Line, the CIMGEN - MANCHACA 138 kV loads to 113.3% of its 391 MVA rating. Project DFAX: 40.2%.

NORWAY PT HYDROPOWER PROJECT - BEARDSLEY 138 kV: 112.4% loading (968 MVA) following the outage of the COLLIERVILLE - JAMES E. ROGERS ENERGY COMPLEX 138 kV; DFAX = 100.0%.

JAMES E. ROGERS ENERGY COMPLEX - COLLIERVILLE 138 kV L: 112.2% loading (975 MVA) following the outage of the JAMES E. ROGERS ENERGY COMPLEX - NORWAY PT HYDROPOWER PROJECT 138 kV line; DFAX = 100.0%.

JAMES E. ROGERS ENERGY COMPLEX - NORWAY POINT HYDROPOWER PROJECT 138 kV line: 112.2% loading (975 MVA) following the outage of the COLLIERVILLE - JAMES E. ROGERS ENERGY COMPLEX 138 kV Line; DFAX = 100.0%.

The MUNISING - COLORADO RIVER SUBSTATION 138 kV is overloaded to 111.6% of its 195 MVA emergency rating for the loss of the N ALLEGHENY WINDPOWER PROJECT SUBSTATION - BAKEOVEN SUBSTATION 138 kV L. The project contributes 26.5% DFAX (65.2 MW) to this overload.

FAIR OAKS 345/138kV Transformer Bank: 110.6% loading (492 MVA) following the outage of the NORWAY POINT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX SUBSTATION 138 kV; DFAX = 45.4%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
RAINTREE - BAKEOVEN SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$1,825,000	22% (\$399,000)
NOTTELY - MARION S 138 kV	Rebuild line with higher-capacity conductor	\$7,721,000	21% (\$1,627,000)
7TH AVENUE N - NOTTELY 138 kV Line	Rebuild line with higher-capacity conductor	\$14,809,000	21% (\$3,121,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
RAINTREE - MUNISING 138 kV L	Rebuild line with higher-capacity conductor	\$2,875,000	22% (\$629,000)
CARL BAILEY - OTSEGO SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$7,334,000	22% (\$1,652,000)
CARREL 345/138 kV TX	Replace transformer with higher-rated unit	\$21,067,000	22% (\$4,742,000)
SPINDLE 500-345 kV Transformer	Replace transformer with higher-rated unit	\$37,246,000	3% (\$1,161,000)
HIGHLAND WIND PROJECT PA - KENNECOTT 138 kV L	Rebuild line with higher-capacity conductor	\$1,793,000	22% (\$402,000)
ALCONA - HIGHLAND WIND PROJECT PA 138 kV Line	Rebuild line with higher-capacity conductor	\$2,809,000	22% (\$631,000)
BARNUM - ALCONA 138 kV	Rebuild line with higher-capacity conductor	\$6,320,000	22% (\$1,419,000)
BARNUM - WOLFSKILL TAP 138 kV Line	Rebuild line with higher-capacity conductor	\$3,612,000	22% (\$811,000)
MANCHACA - O'BANION 138 kV	Rebuild line with higher-capacity conductor	\$7,488,000	22% (\$1,637,000)
WOLFSKILL TAP - MERRILL HILLS SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$4,751,000	22% (\$1,067,000)
COLLIERVILLE - MERRILL HILLS 138 kV	Rebuild line with higher-capacity conductor	\$6,155,000	22% (\$1,382,000)
O'BANION - PURDUE NORTHWEST SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$5,777,000	22% (\$1,263,000)
BEARDSLEY - PURDUE NORTHWEST 138 kV	Rebuild line with higher-capacity conductor	\$4,450,000	22% (\$973,000)
N ALLEGHENY WINDPOWER PROJECT - BAKEOVEN 138 kV	Rebuild line with higher-capacity conductor	\$1,804,000	22% (\$397,000)
BEARDSLEY - BAKEVEN 138 kV line	Rebuild line with higher-capacity conductor	\$15,312,000	22% (\$3,366,000)
7TH AENUE N - CIMGEN 138 kV Line	Rebuild line with higher-capacity conductor	\$3,596,000	21% (\$757,000)
MANCHACA - CIMGEN 138 kV	Rebuild line with higher-capacity conductor	\$1,602,000	21% (\$337,000)
NORWAY PT HYDROPOWER PROJECT - BEARDSLEY 138 kV	Rebuild line with higher-capacity conductor	\$2,815,000	22% (\$632,000)
JAMES E. ROGERS ENERGY COMPLEX - COLLIERILLE 138 kV L	Rebuild line with higher-capacity conductor	\$4,195,000	22% (\$942,000)
NORWAY PT HYDROPOWER PROJECT - JAMES E. ROGERS ENERGY COMPLEX 138 kV L	Rebuild line with higher-capacity conductor	\$11,789,000	22% (\$2,647,000)
MUNISING - COLORADO RIVER SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$2,346,000	22% (\$513,000)
FAIR OAKS 345/138kV Transformer Bank	Replace transformer with higher-rated unit	\$15,419,000	22% (\$3,462,000)
	Total allocated to project		\$35,969,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.